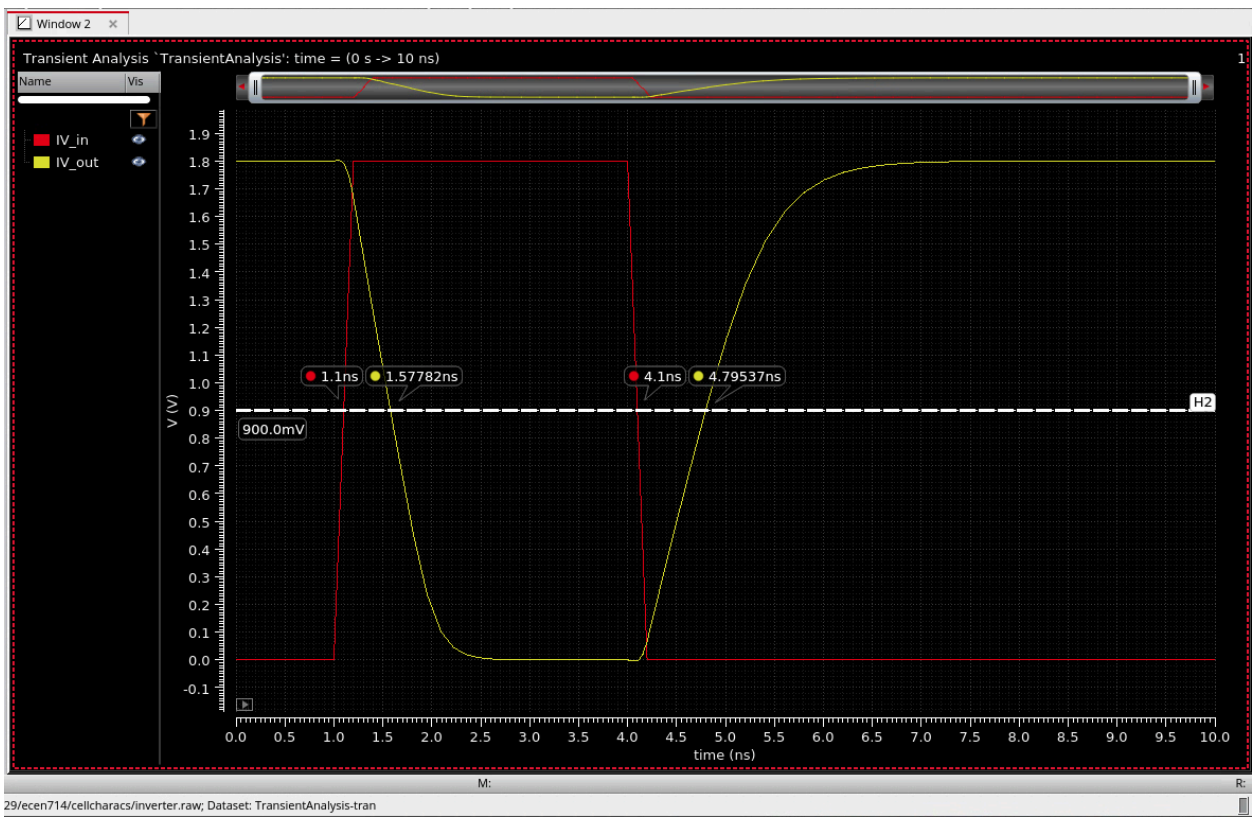
ECEN 714: DIGITAL INTEGRATED CIRCUIT DESIGN
LAB REPORT - 3

BY: YASHAS BHARADWAJ (UIN - 635009332)

Inverter

1. Waveform:

PMOS: Width = 1.2um & Length = 0.2um
NMOS: Width = 0.4um & Length = 0.2um



2. Cell Delay Table:

The delay is calculated as for example **falling delay = 1.1 - 1.1284** for 1fF load capacitance value. Similarly for the rising delay it is subtracted from **4.1**.

Sr. No.	Load Capacitance (fF)	Falling Delay (ns)	Rising Delay (ns)	Error%
1	1	0.0284	0.0419	47.54
2	7	0.0645	0.07567	17.32
3	15	0.09963	0.1065	6.90
4	21	0.1212	0.1263	4.21
5	28	0.1466	0.1494	1.91
6	34	0.1684	0.1691	0.42
7	41	0.1938	0.1926	0.62
8	48	0.2187	0.2165	1.01
9	54	0.2402	0.2360	1.75
10	60	0.2615	0.2580	1.34
11	67	0.2871	0.2812	2.06
12	74	0.312	0.3029	2.92
13	80	0.3351	0.3251	2.98
14	86	0.3571	0.3466	2.94
15	100	0.4063	0.3933	3.20

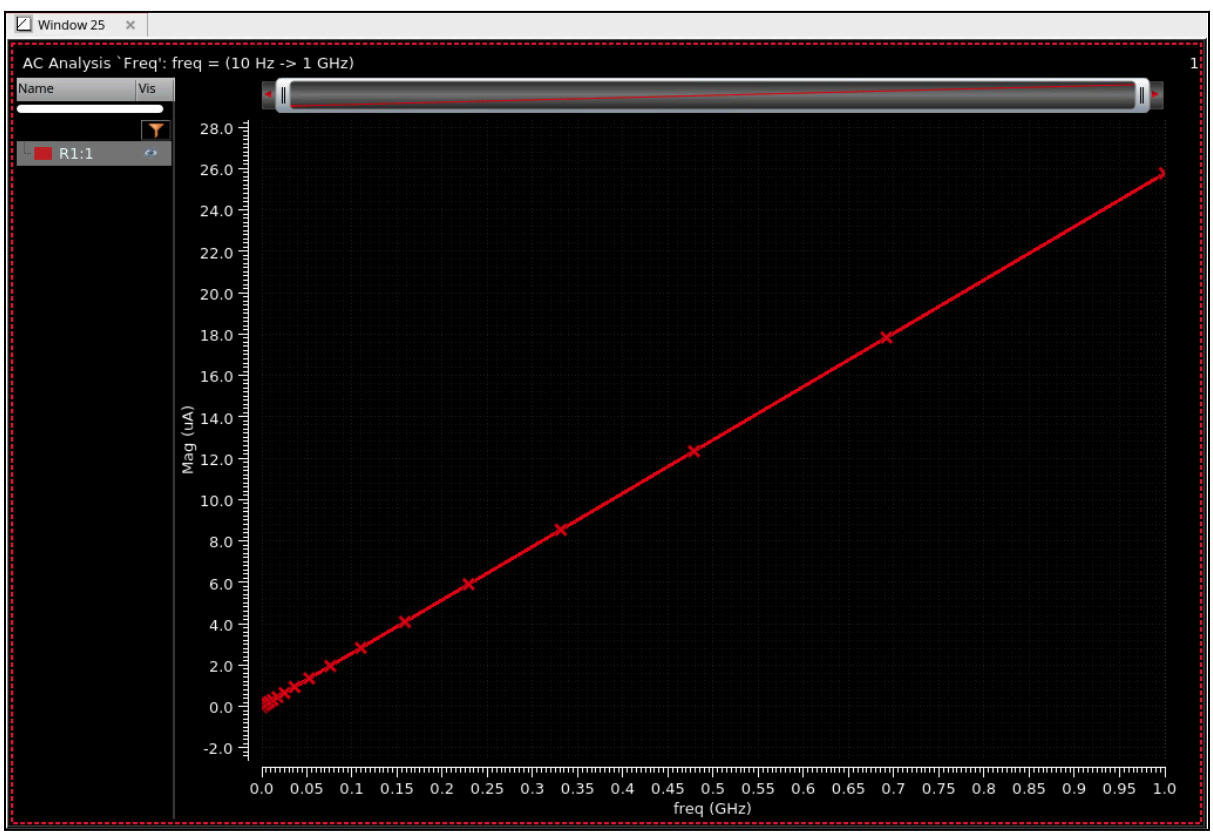
As it can be seen from the table above, when C = 100f, the falling delay = 0.406ns and rising delay = 0.393ns. The difference is 11ps. Therefore with respect to the falling delay, the error is 3.2%.

3. Sink Capacitance:

The sink capacitance is calculated as $\mathbf{C = I/2\pi f}$.

Serial Number	Frequency (MHz)	Current (uA)	Sink Capacitance (fF)
1	36.3	0.935	4.100
2	52.48	1.35	4.094
3	75.85	1.95	4.092
4	109	2.8	4.089
5	158	4.08	4.110
6	229	5.9	4.101
7	331	8.5	4.087
8	478	12.33	4.106
9	691	17.8	4.100
10	1000	25.76	4.100

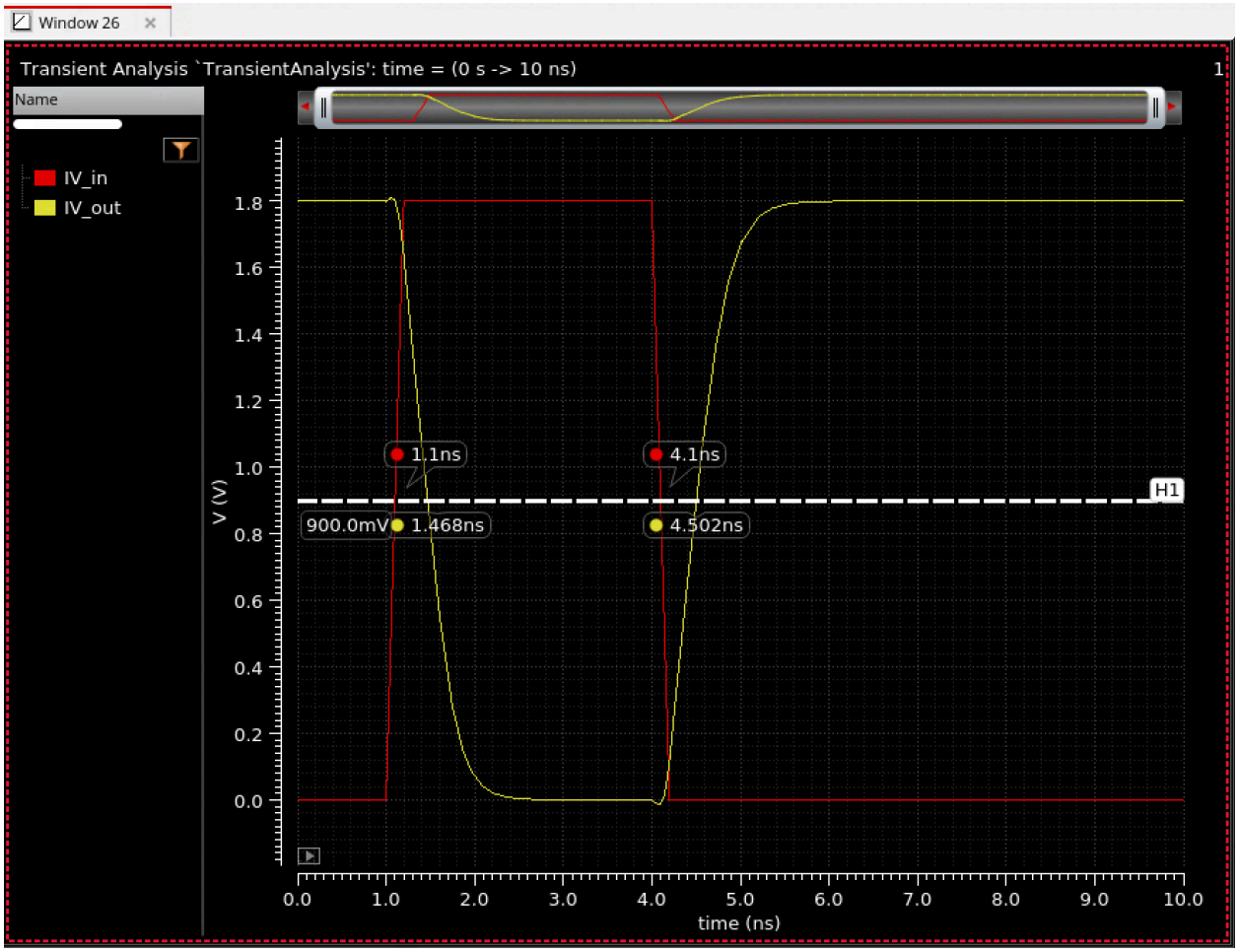
The mean of all the sink capacitances is **4.09 fF**.



NAND2

1. Waveform:

PMOS: Width = 1.2um & Length = 0.2um
NMOS: Width = 0.75um & Length = 0.2um



2. Cell Delay Table:

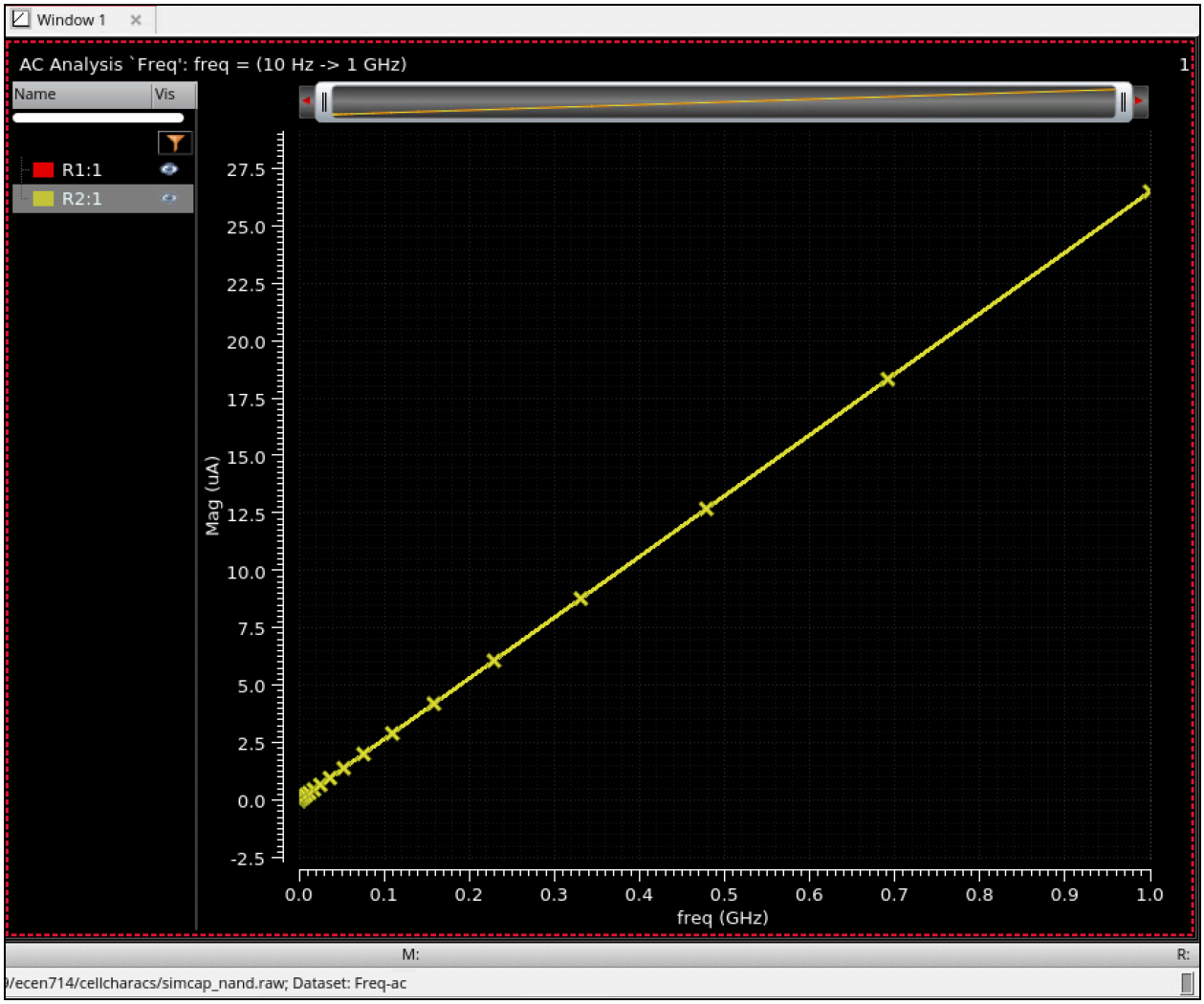
Sr. No.	Load Capacitance (fF)	Falling Delay (ns)	Rising Delay (ns)	Error%
1	1	0.034	0.053	55.88
2	7	0.065	0.084	29.23
3	15	0.096	0.113	17.71
4	21	0.115	0.133	15.65
5	28	0.138	0.157	13.77
6	34	0.156	0.177	13.46
7	41	0.179	0.201	12.29
8	48	0.201	0.225	11.94
9	54	0.221	0.245	10.86
10	60	0.24	0.266	10.83
11	67	0.261	0.289	10.73
12	74	0.283	0.312	10.25
13	80	0.304	0.335	10.20
14	86	0.322	0.354	9.94
15	100	0.368	0.402	9.24

As shown in the table, when C = 100fF, the falling delay = 368ns and rising delay = 402ns. The error is calculated as **9.24%** wrt the falling delay.

3. Sink Capacitance:

Serial Number	Frequency (MHz)	Current (uA)	Sink Capacitance (fF)
1	36.3	0.961	4.214
2	52.48	1.39	4.216
3	75.85	2	4.197
4	109	2.9	4.235
5	158	4.196	4.227
6	229	6.06	4.212
7	331	8.76	4.212
8	478	12.67	4.219
9	691.83	18.318	4.214
10	1000	26.47	4.213

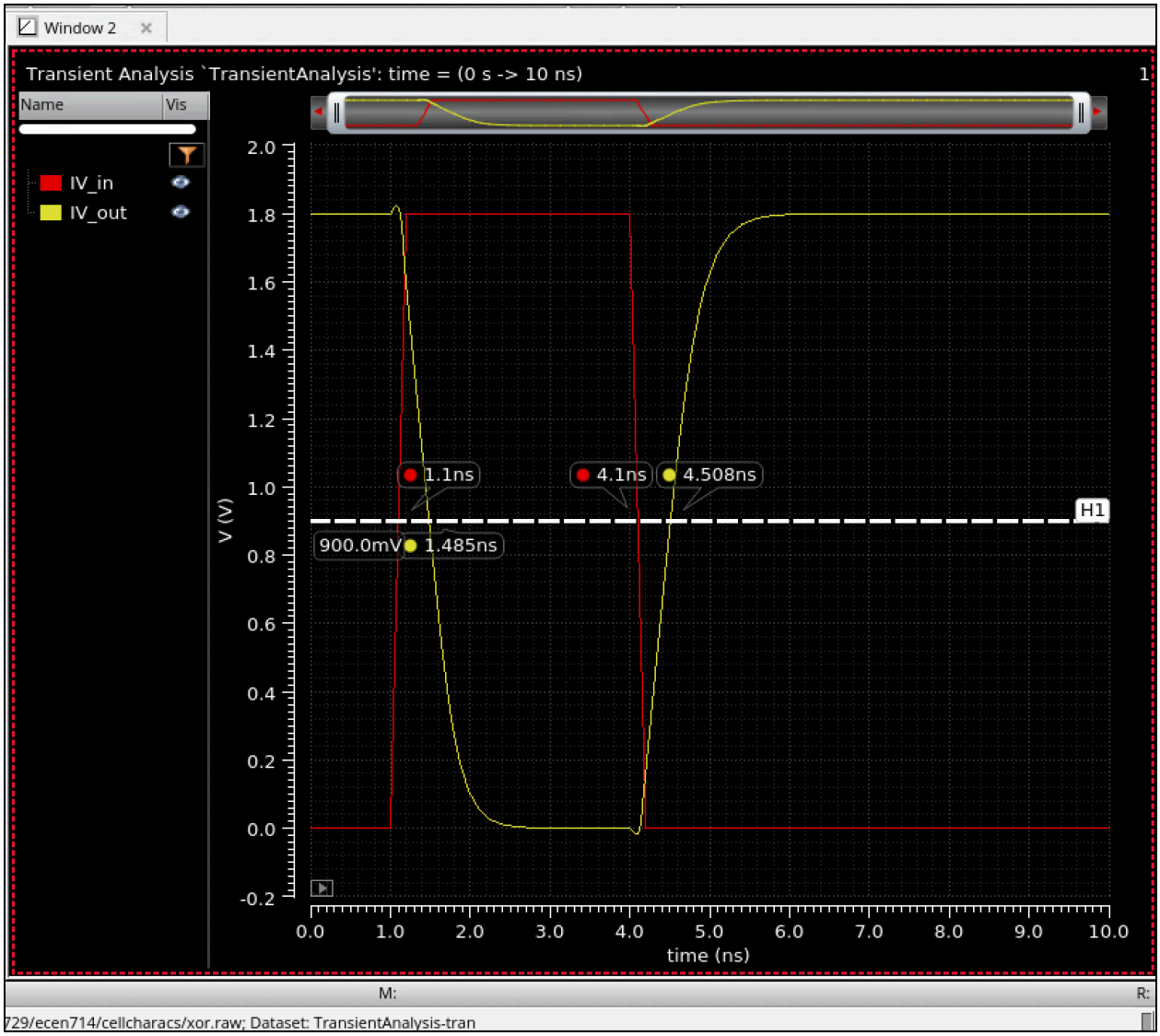
The mean of all sink capacitances is **4.21fF**.


Window 1
AC Analysis `Freq': freq = (10 Hz -> 1 GHz)
Name
Vis
R1:1
R2:1
Mag (uA)
27.5
25.0
22.5
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
-2.5
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
freq (GHz)
M:
R:
/ecen714/cellcharacs/simcap_nand.raw; Dataset: Freq-ac

XOR2

1. Waveform:

PMOS: Width = 2.2um & Length = 0.2um
NMOS: Width = 0.7um & Length = 0.2um

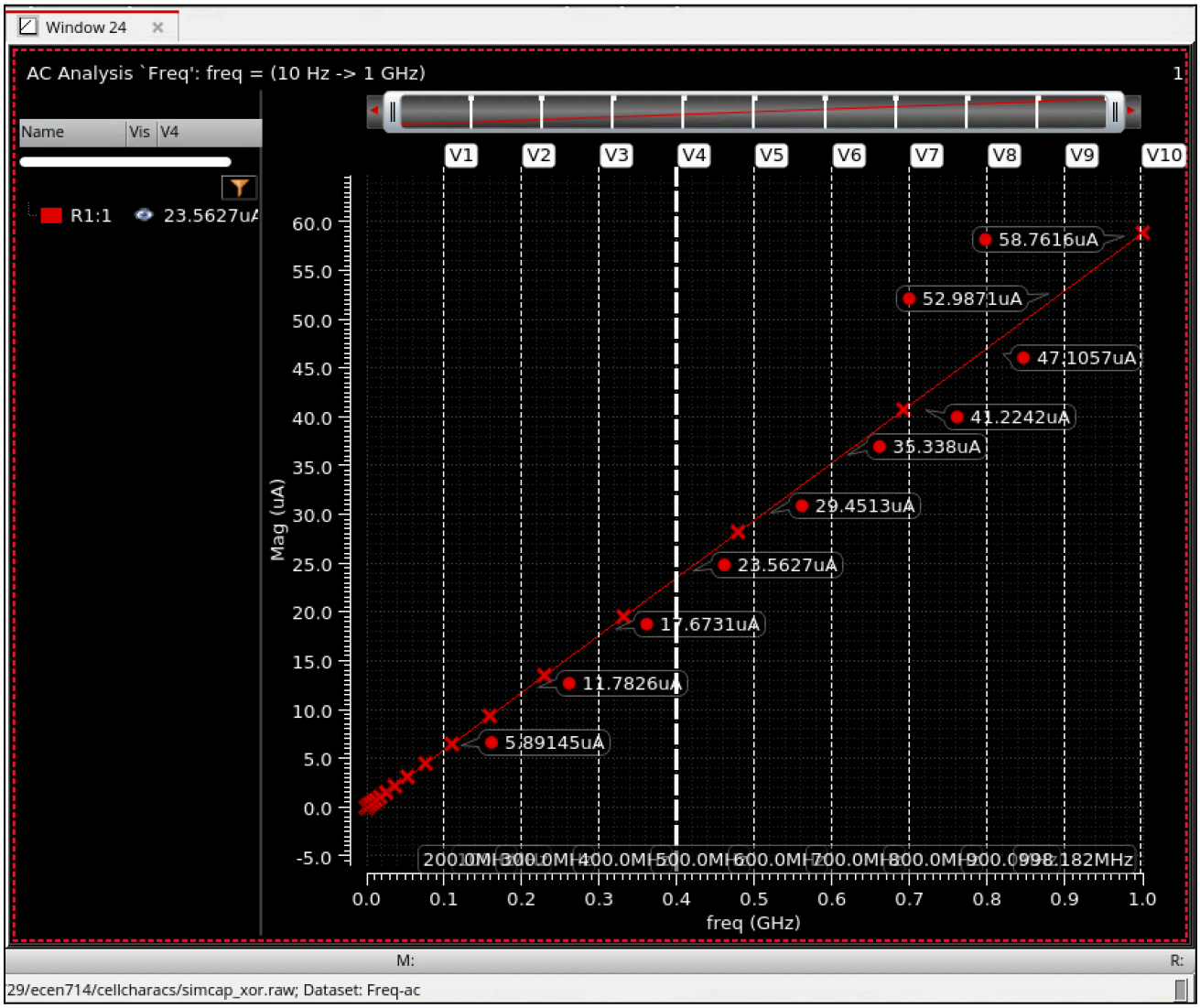


2. Cell Delay Table:

Sr. No.	Load Capacitance (fF)	Falling Delay (ns)	Rising Delay (ns)	Error%
1	1	0.041	0.048	17.07
2	7	0.066	0.073	10.61
3	15	0.096	0.105	9.37
4	21	0.116	0.126	8.62
5	28	0.14	0.151	7.86
6	34	0.161	0.173	7.45
7	41	0.185	0.198	7.03
8	48	0.209	0.223	6.70
9	54	0.229	0.245	6.99
10	60	0.25	0.266	6.40
11	67	0.274	0.291	6.20
12	74	0.297	0.316	6.40
13	80	0.318	0.337	5.97
14	86	0.338	0.36	6.51
15	100	0.385	0.408	5.97

Hence the delay error% for 100fF is **5.97%** i.e. below 10%.

3. Sink Capacitance:



Serial Number	Frequency (MHz)	Current (uA)	Sink Capacitance (fF)
1	100	5.89	9.375
2	200	11.78	9.375
3	300	17.67	9.375
4	400	23.56	9.375
5	500	29.45	9.375
6	600	35.33	9.372
7	700	41.22	9.372
8	800	47.1	9.371
9	900	52.99	9.371
10	998	58.76	9.371

The mean of all the capacitances is **9.372fF**.

Simulation Files

Cell18.spi file (Same for all circuits)

MobaTextEditor

File Edit Search View Format Syntax Special Tools

nand.spi | cell18.spi | xor.spi | xor_delaytable.spi | simcap_xor.spi | simcap_nand.spi | inverter.spi

```
//Spice netlist for an inverter
;By Yashas Bharadwaj 635009332
simulator lang=spectre
subckt IV (input output VDD VSS)
        parameters wp=0.6u lp=0.2u wn=0.3u ln=0.3u
        M1 output input VDD VDD tsmc18P w=wp l=lp
        M2 output input VSS VSS tsmc18N w=wn l=ln
ends IV

subckt NAND2 (inputA inputB output VDD VSS)
        parameters wp=1.2u lp=0.2u wn=0.75u ln=0.2u
        M1 output inputA VDD VDD tsmc18P w=wp l=lp
        M2 output inputB VDD VDD tsmc18P w=wp l=lp
        M3 output inputA 1 VSS tsmc18N w=wn l=ln
        M4 1 inputB VSS VSS tsmc18N w=wn l=ln;
ends NAND2

subckt XOR2 (inputA inputB output VDD VSS)
        parameters wp=1.2u lp=0.2u wn=0.75u ln=0.2u

        IV1 (inputA Abar VDD VSS) IV wp=wp lp=lp wn=wn ln=ln
        IV2 (inputB Bbar VDD VSS) IV wp=wp lp=lp wn=wn ln=ln

        M1 1 inputB VDD VDD tsmc18P w=wp l=lp
        M2 output Abar 1 VDD tsmc18P w=wp l=lp
        M3 2 Bbar VDD VDD tsmc18P w=wp l=lp
        M4 output inputA 2 VDD tsmc18P w=wp l=lp
        M5 output Abar 3 VSS tsmc18N w=wn l=ln
        M6 3 Bbar VSS VSS tsmc18N w=wn l=ln
        M7 output inputA 4 VSS tsmc18N w=wn l=ln
        M8 4 inputB VSS VSS tsmc18N w=wn l=ln
ends XOR2
```

* C:\Users\YASHAS~1\AppData\Roaming\MobaXterm\slash\Remot UNIX | Plain text | 39 lines | Row #2 | Col #31

INVERTER

1. inverter.spi file

```
;Spice netlist for an inverter and a capacitor
; By Yashas Bharadwaj 635009332
simulator lang=spectre

include "~/ecen714/cellcharacs/model18.spi"
include "~/ecen714/cellcharacs/cell18.spi"

vgnd (gnd 0) vsource dc=0
vvdd (vdd 0) vsource dc=1.8

vpwl (IV_in 0) vsource type=pwl wave=[0n 0 1n 0 1.2n 1.8 4n 1.8 4.2n 0]

X1 (IV_in IV_out vdd gnd) IV wp=1.2u lp=0.2u wn=0.4u ln=0.2u

R1 (IV_out 1) resistor r=1
C1 (1 0) capacitor c=100f

TransientAnalysis tran start=0 stop=10ns step=1ps
save IV_in IV_out
```

2. simcap_inverter.spi file

```
;Spice netlist for an inverter and a capacitor
simulator lang=spectre

include "~/ecen454_714/cellcharacs/model18.spi"
include "~/ecen454_714/cellcharacs/cell18.spi"

vgnd (gnd 0) vsource dc=0
vvdd (vdd 0) vsource dc=1.8

vpwl (IV_in 0) vsource type=pwl wave=[0n 0 1n 0 1.2n 1.8 4n 1.8 4.2n 0]

X1 (IV_in IV_out vdd gnd) IV wp=0.6u lp=0.2u wn=0.3u ln=0.2u

R1 (IV_out 1) resistor r=1
C1 (1 0) capacitor c=100f

TransientAnalysis tran start=0 stop=10ns step=1ps
save IV_in IV_out
```

NAND2

1. nand.spi file

```
;Spice netlist for an nand and a capacitor
;By Yashas Bharadwaj 635009332
simulator lang=spectre

include "~/ecen714/cellcharacs/model18.spi"
include "~/ecen714/cellcharacs/cell18.spi"

vgnd (gnd 0) vsource dc=0
vvdd (vdd 0) vsource dc=1.8

vpwl (IV_in 0) vsource type=pwl wave=[0n 0 1n 0 1.2n 1.8 4n 1.8 4.2n 0]
N1 (IV_in vdd IV_out vdd gnd) NAND2 wp=1.2u lp=0.2u wn=0.75u ln=0.2u

R1 (IV_out 1) resistor r=1
C1 (IV_out 0) capacitor c=100f

TransientAnalysis tran start=0 stop=10ns step=1ps
save IV_in IV_out
```

2. simcap_nand.spi file

```
;Spice netlist for an inverter and a capacitor
;By Yashas Bharadwaj 635009332
simulator lang=spectre

include "~/ecen714/cellcharacs/model18.spi"
include "~/ecen714/cellcharacs/cell18.spi"

vgnd (gnd 0) vsource dc=0
vvdd (vdd 0) vsource dc=1.8

acinputA (A 0) vsource dc=0 mag=1
acinputB (B 0) vsource dc=0 mag=1

R1 (A A1) resistor r=0
R2 (B B1) resistor r=0

X1 (A1 B1 NAND2_out vdd gnd) NAND2 wp=1.2u lp=0.2u wn=0.75u ln=0.2u

Freq ac start=1e+1 stop=1e+9
save R1:currents R2:currents
```

XOR2

1. xor.spi file

```
;Spice netlist for an nand and a capacitor
;By Yashas Bharadwaj 635009332
simulator lang=spectre

include "~/ecen714/cellcharacs/model18.spi"
include "~/ecen714/cellcharacs/cell18.spi"

vgnd (gnd 0) vsource dc=0
vvdd (vdd 0) vsource dc=1.8

vpwl (IV_in 0) vsource type=pwl wave=[0n 0 1n 0 1.2n 1.8 4n 1.8 4.2n 0]

XOR1 (IV_in vdd IV_out vdd gnd) XOR2 wp=2.2u lp=0.2u wn=0.7u ln=0.2u

R1 (IV_out 1) resistor r=1
C1 (IV_out 0) capacitor c=100f

TransientAnalysis tran start=0 stop=10ns step=1ps
save IV_in IV_out
```

2. simcap_xor.spi file

```
;Spice netlist for an inverter and a capacitor
;By Yashas Bharadwaj 635009332
simulator lang=spectre

include "~/ecen714/cellcharacs/model18.spi"
include "~/ecen714/cellcharacs/cell18.spi"

vgnd (gnd 0) vsource dc=0
vvdd (vdd 0) vsource dc=1.8

acinput (XOR2_in 0) vsource dc=0 mag=1

R1 (XOR2_in XOR2_in1) resistor r=0

X1 (vdd XOR2_in1 XOR2_out vdd gnd) XOR2 wp=1.2u lp=0.2u wn=0.75u ln=0.2u


Freq ac start=1e+1 stop=1e+9
save R1:currents
```